



**PORTABLE AIR
CONDITIONERS**

INDEX

1. GENERAL INFORMATION	4
1.1. About Units	4
1.1.1. Optional Properties.....	4
1.2. Smart Code	5
1.3. Unit Label.....	6
1.4. Warranty Conditions	7
1.5. IP54 Rating	7
1.5.1. First Digit - Protection Against Solid Objects	9
1.5.2. Second Digit - Protection Against Liquids	9
1.6. Refrigerant	10
1.6.1. Safety Instructions for Maintenance	10
1.6.2. Equipment Used in Maintenance	11
1.6.3. Transport.....	12
1.6.4. Storage and Installation	13
2. TECHNICAL INFORMATION.....	14
2.1. Technical Information.....	14
2.2. Cooling Cycle.....	15
2.3. Equipment Information	16
2.4. Dimensions and Weight	17
2.5. Minimum Maintenance Clearance	18
2.6. Unit Drawings	18
3. CHECK LIST	19
3.1. Delivery Options.....	19
3.2. Before Installation	19
3.3. Connections.....	19
3.4. Before Running	20
4. CARRYING and STORAGE	21
4.1. Considerations During Transportation	21
4.2. Carrying with Crane or Crane-like Vehicles	21
4.3. Carrying with Forklift or Forklift-like Vehicle.....	21
4.4. Unpacking	21
4.5. Considerations During Storage.....	23
5. INSTALLATION	24
5.1. About Installation.....	24
5.2. Electrical connection	24
6. COMMISSIONING	25
6.1. Control Panel	25
6.1.1. Control Panel Technical Dimensions	25

6.1.2.	Control Panel Technical Specifications.....	26
6.1.3.	Panel Guide.....	27
6.1.4.	Weekly Schedule Configuration.....	27
6.1.5.	Date and Hour Configuration.....	28
6.1.6.	Language Configuration.....	28
6.1.7.	Display Brightness Configuration.....	28
6.1.8.	Screen Contrast Configuration.....	28
6.2.	Service Configuration.....	29
6.2.1.	Temperature Sensor Position Setting.....	29
6.2.2.	Defrost Service Setting.....	29
6.2.3.	Outdoor Temperature Display.....	29
6.2.4.	Minimum Set Temperature Setting.....	29
6.2.5.	Maximum Set Temperature Setting.....	29
6.2.6.	Automatic Mode Minimum Fan Speed Temperature.....	30
6.2.7.	Condenser Set Temperature Value Setting.....	30
6.2.8.	Compressor Start Time Setting.....	30
6.2.9.	Compressor Stop Time Setting.....	30
6.2.10.	Panel Communication Setting.....	30
6.2.11.	BMS Modbus Communication Setting.....	30
6.2.12.	Panel Factory Reset.....	31
6.3.	Modbus List.....	32
7.	MAINTENANCE.....	37
7.1.	Information for Using the Unit.....	37
7.2.	Condenser Cleaning.....	37
7.3.	Compressor Oil Check.....	37
7.4.	Leakage Check.....	37
7.5.	Fan Check.....	38
7.6.	Prob Check.....	38
7.7.	Refrigerant Check.....	38
7.8.	Filter Cleaning.....	38
8.	ERRORS and WARNINGS.....	39
9.	SPARE PART LIST.....	40

INTRODUCTION

The aim of this document is to guide the user through installation and maintenance.

Read the instructions carefully. The installation and maintenance of the unit should only be carried out by authorized personnel. The warranty is cancelled if the user modifies the unit in any way. Follow the local law when installing the unit. This document should be stored for future use. The packaging should be kept away from children and should be recycled as much as possible. All units are designed to function outdoors. If the unit has taken damage or is not functioning properly, turn off the main on/off switch. If the unit includes an electric heater, turn off the IAM switch located on the electric heater. Please contact the authorized service and use original parts in any case of error. Please follow instructions thoroughly for maximum safety. Producer company cannot be held responsible for errors if the user does not follow instructions.

Helot reserves the right to change the manual information without notifying the buyer.

1. GENERAL INFORMATION

1.1. About Units

- Automatic defrost
- R-454B refrigerant,
- Scroll compressor,
- Heating,
- Cooling,
- Cooling + heat pump,
- LCD display,
- Filter alarm,
- EC Plug fan,
- Electronic expansion valve,
- Tam automatic,
- Plug and run,
- Easy service and maintenance,
- Portative.

Helot portable roof type air conditioners can be installed indoors and outdoors. Reinforced carcass and special body structure for easy installation and portability are designed for use in various places. Larger capacity needs are met with multiples of the unit.

In TSE 121 model suitable for outdoor environment; In the condenser of the unit, axial type, self-powered, silent and high-efficiency fans, optionally speed-controlled and EC plug fans are used.

In TRT 121 model suitable for indoor environment; In the condenser of the unit, radial AC plug fan, silent and high efficiency fans with direct coupled motor, optionally speed controlled and EC plug fans are used.

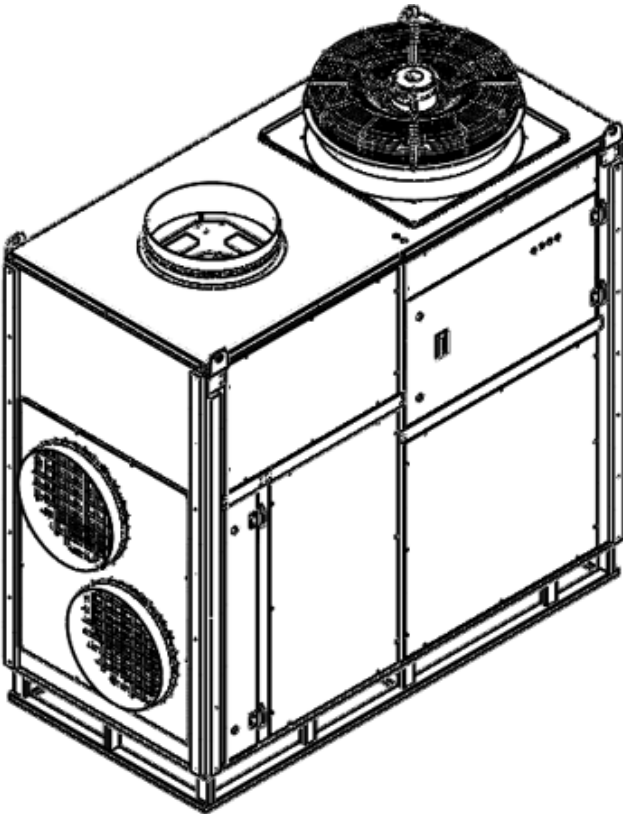
Helot portable rooftop air conditioners are designed to provide ideal climatic conditions for indoor and outdoor use. Helot portable roof type air conditioner, which responds to all the air conditioning needs for certain periods of time in the service sector, events and corporate organizations with its effective performance and affordable cost, offers a portable solution in air conditioning.

The units have the ability to operate in a wide temperature range (-10 °C to +45 °C). R-454B refrigerant, which does not harm the ozone layer, is used as the refrigerant.

1.1.1. Optional Properties

- Remote control with PC or building automation,
- Copper tube, anodized or hydrophilic aluminum finned condenser,
- copper tube, copper fin condenser,
- Heating with electric resistance,
- Compressor with inverter,
- Flexibility in designs and models suitable for special projects.

Figure 1: Isometric view



1.2. Smart Code

Figure 2: Smart Code

RT	H	G	M	016	2
↓	↓	↓	↓	↓	↓
Rooftop	Air Cooled	Portable	R-454B Refrigerant	Compresor Nominal Power	Compressor Quantity

1.3. Unit Label

The label includes cooling capacity, heating capacity, refrigerant information, operating voltage and current information, and storage conditions.

Figure 3: Example unit label

	
CE EAC TSEK	
PORTABLE AIR CONDITIONERS	
SERIAL NUMBER 2500106-00943-17	
MODEL HSE.0121	PRODUCTION YEAR 2025
PRODUCTION CODE RT.HGM.0121	REFRIGERANT R454B
OPERATING VOLTAGE 380~420V/3Ph/50Hz	REFRIGERANT CHARGE 4,85 kg
ÜRETİCİ / MADE BY : İMBAT SOĞUTMA ISITMA MAKİNA SANAYİ VE TİCARET A.Ş. İZMİR - TÜRKİYE TEL: + 90 232 877 21 01 FAX: + 90 232 877 21 04 www.imbat.com	

1.4. Warranty Conditions

- 1) Except for the following points, the entire product, including all parts, is under the guarantee of our company;
 - a) Deterioration, breakage or scratching of the outer surface of the product,
 - b) Damages and errors that occur during loading, unloading and transportation,
 - c) Damages and errors caused by low or excess voltage, faulty electrical installation, usage reasons different from the voltage written on the plate of the unit,
 - d) Failures arising from the replacement of the components of the unit with any other than Helot standards,
 - e) Opening the product for maintenance, repair or any reason other than the service personnel authorized by Helot, causing damage or error,
 - f) Damages and errors that occur due to the observance of the items specified in the user manual of the product,
- 2) The warranty period is two (2) years from the invoice date.
- 3) If the product fails during the warranty period, the time spent in repair is added to the warranty period. The repair period of the product is one (1) month at the most. For a while, the product will be sent to the service station, in the absence of a service station, respectively; It starts from the date the product is delivered to one of the seller, dealer, agency, representative, importer or manufacturer.
- 4) If the product fails during the warranty period due to material and workmanship or assembly errors; It will be repaired without any charge due to labor cost, replacement part cost or any other reason.
- 5) The product will be exchanged free of charge in the following cases:
 - a) Within one year from the delivery date, provided that it remains within the warranty period; The continuity of not being able to benefit from the product as a result of repeating the same error more than four times or different errors occurring more than six times,
 - b) Exceeding the maximum time required for repair,
 - c) In case the service station is not available, respectively; Determining that the repair of the fault is not possible with a report to be issued by one of the seller, dealer, agency, representative, importer or manufacturer of the product.

Certificate of Guarantee must be presented when requested.

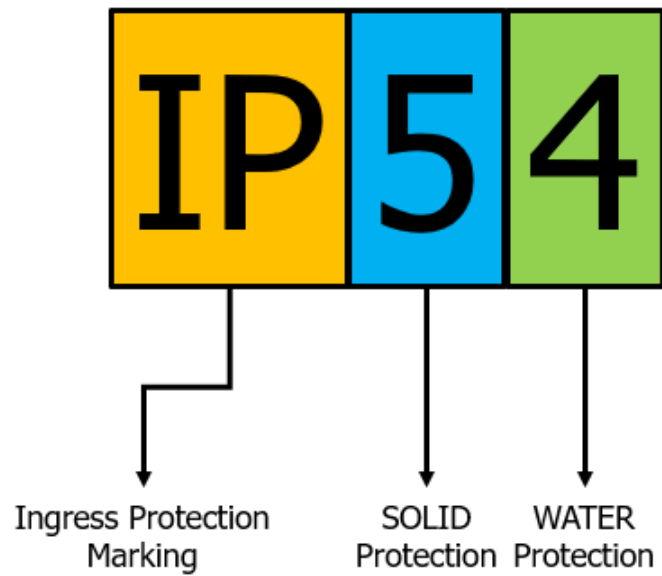
1.5. IP54 Rating

The degree of protection that an object has against solid objects such as dust, sand and dirt and liquids is called IP Protection Class. IP protection codes that can be described as universal; CENELEC, headquartered in Brussels, has been determined according to the standard numbered IEC-60529 published by the European Committee for Electromechanical Standardization and has been accepted by the whole world. The IP rating consists of a 2 or 3 digit number; 3rd digit is mostly not used because it is not included in IEC 60529.

Digits Used in IP Classification and Their Meanings:

1. Protection against solid objects
2. Protection against liquids
3. Protection from mechanical effects (it is not used very often because it is not included in the standard in IEC 60529).

Figure 4: IP Classification and Their Meanings



1.5.1. First Digit - Protection Against Solid Objects

- 1) Protection against solid objects up to 5 cm (50 mm) in diameter: Protection against access by a thick stick, pipe and objects of approximately this thickness to dangerous points. Objects larger than 5 cm in diameter cannot touch the inside of the unit.
- 2) Protection against solid objects 1.25 cm (12.5 mm) in diameter: Protection against access to dangerous points Fingers, the back of a fountain pen... and objects of approximately this thickness. Objects larger than 1.25 cm in diameter cannot touch the inside of the unit.
- 3) Protection against solid objects 0.25 cm (2.5 mm) in diameter: Protection against access to dangerous points by the male entry end of an average cable, screwdriver and objects of approximately this thickness. Objects larger than 0.25 cm in diameter cannot touch the inside of the device.
- 4) Protection against solid object 0.1 cm (1 mm) in diameter: Protection against access to dangerous points by pen tip, wire and objects of approximately similar thickness. Objects larger than 0.1 cm in diameter cannot touch the inside of the device.
- 5) Protection against dust: Reducing dust ingress to hazardous points to levels that are not harmful. Dust cannot be completely avoided. It is enough to prevent the dust from damaging the product.
- 6) Dustproof: The ingress of dust is completely blocked. Dust is prevented from damaging the product.

1.5.2. Second Digit - Protection Against Liquids

- 1) Protection against vertically (Vertical) water drops: Provides protection against vertically (Vertical) dripping water on the device.
- 2) Protection against drops of water falling on the enclosure at an oblique angle of up to 15 degrees: Protection against water flow, such as rain in windy weather. It provides protection from water drops at an angle of 0 to 15 degrees to the device.
- 3) Protection against water jets: Protection against incident water jets such as fire alarm system water, sprinklers, etc. Water sprayed at an angle of 60 degrees does not damage the device.
- 4) Protection against splashes: from the ceiling, splashed from the floor or by the wind; protection against splashing water from all directions. Water splashing from any direction will not damage the device.
- 5) Protection against splashing water: Protection against water jets from a small hose. Low pressure water coming from any direction will not damage the device.
- 6) Protection against strong jets of water: Protection against high-intensity water jets, such as a fire hose. High pressure water coming from any direction will not damage the device.
- 7) Protection against immersion in water for less than 30 minutes: Protection against immersion or short-term flooding. Provides maximum 30 minutes of protection in water at the specified pressure and temperature.
- 8) Protection against the effects of immersion in water: It is the most water-resistant protection standard. It provides the ability to work for a long time in water under the specified temperature and pressure.

1.6. Refrigerant

A2L Class Refrigerants

Non-toxic and slightly flammable A2L refrigerants offer a relatively easy solution to reduce global warming potential. A2L refrigerants provide better performance and efficiency than many A1 and natural refrigerants, making them suitable for versatile choices with the right equipment. The place of A2L fluids among the refrigerant classes in ASHRAE34/ISO817 standards is shown in the table below.

A = Non-Toxic
2 = Flammable
L = Low Burning Rate

ISO and IEC standards are used as international security standards. The relationships between international and European standards are shown in the tables below:

Table 1: ISO and IEC standards

Standard Type		International	Europe
Refrigerant Classification		ISO 817	Follows ISO 817
Safety Regulation in Use	General	ISO 5149	EN378
	Equipment Specific	IEC 60335-2-24	EN 60335-2-24
		IEC 60335-2-40	EN 60335-2-40
		IEC 60335-2-89	EN 60335-2-89

R-454B, which has a lower KIP level compared to refrigerants such as R-134a and R-410A, is in the A2L refrigerant class. If we take a look at the R-410A – R-32 alternatives used in the industry, the R-454B fluid used by Portable Air Conditioners has a lower KIP level than its counterparts.

Table 2: Flammability

Flammability	Low Toxicity	High Toxicity
No Flame Spread	A1: CFC, HCFC, many HFCs	B1
Low Flammability	A2L: Many HFOs, R-32, R-454B	B2L: Ammonia
flammability	A2: R152	B2
High Flammability	A3: Hydrocarbons	B3

1.6.1. Safety Instructions for Maintenance

- Follow the information provided in EN 378:2016-2019.
- R-454B is heavier than air and accumulates on the ground in case of leakage. Accumulation of refrigerant in closed rooms (warehouse) creates explosion hazard and suffocating atmosphere. Make sure that the atmosphere where maintenance is performed is adequately ventilated.
- While performing installation and maintenance, keep it away from ignition sources such as gas burners and electric heaters.
- It is not recommended to store and use the unit in ATEX areas.
- Vacuum pumps should be used with a check valve or solenoid valve.
- Only pipe cutters should be used to cut pipes and no torches should be used.
- A leak detector suitable for A2L refrigerants should be used.
- Brazing operations should be carried out in a nitrogen atmosphere.
- There must be fire extinguishing equipment (CO₂ or dry powder) in the environment when performing any operation on the unit.
- Leak detectors must be sufficiently insulated and non-flammable.

- All maintenance personnel must be familiar with the system.
- If compressors or compressor oil are removed, the lubricant must be kept at a certain height to prevent refrigerant from remaining in it.
- The refrigerant should not be released into the atmosphere and should be stored in storage tanks.
- Personnel working on the unit must be qualified and trained on A2L refrigerants systems.

1.6.2. Equipment Used in Maintenance

When using systems containing A2L refrigerants, the equipment used in the table below is used.

Table 3: Equipment Used in Maintenance

Tool	Instructions
Test Valve 	Since the refrigerant filling is done in liquid phase, it is recommended to use a test valve with a sight glass.
Filling Valve 	The fill valve prevents refrigerant from escaping from the hose and air conditioning unit.
Electronic scales 	The exact filling amount is determined with the help of an electronic scale.
Electronic Tightness Tester 	Before operating the electronic leak test device, it must be determined whether it is suitable for the refrigerant.

Vacuum Pump 	Since A2L refrigerants are low flammable, a vacuum pump suitable for this fluid must be used.
Check Valve 	A check valve adapter should be used to prevent oil backflow from the vacuum pump to the fill valve.

1.6.3. Transport

A2L classification is only related to the refrigerant; air conditioning and heat pump system safety is ensured by standards such as EN378, which determine the maximum amount of refrigerant. Units containing A2L refrigerant must comply with ADR (International Transport of Dangerous Goods by Road) rules and UN3358.

If there is more than 12 kg of refrigerant in the unit, the amount of refrigerant must be stated on the shipping receipt. During land transportation, the presence of the following devices in the vehicle protects the driver, vehicle and unit:

- Fire extinguisher
- Explosion proof flashlight
- Red label on unit packaging (Factory)

In the case of maritime transport, IMDG (International Maritime Dangerous Goods Directive) is used, which is similar to ADR rules. Some regions have their own regulations, local laws of the unit destination should be checked.



(No. 2.1)

Flammable gases

Symbol (flame): black or white;
(except as provided in 5.2.2.2.1.6 (d))

Ground: red; Number '2' in the bottom corner

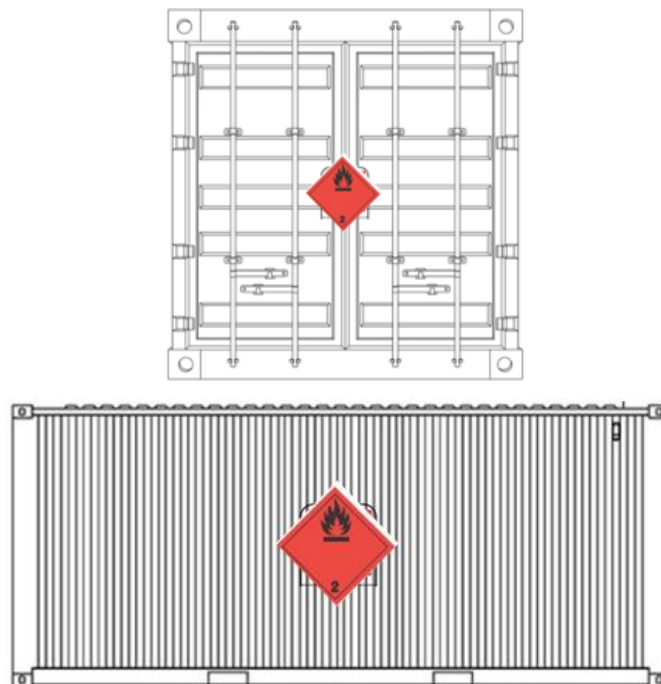


Figure 5: Example label

1.6.4. Storage and Installation

Rooftop units should be installed on rooftops to operate outdoors. For safety reasons, the units must be stored outdoors. If it will be stored indoors before installation, the following conditions are met for the storage building:

- There must be non-flammable material or waste and adequate ventilation.
- Sockets, lamps and switches should be moved away from ignition sources such as electric motors.
- It should be removed from heat sources and should not be directly exposed to the sun.
- Storage area safety signs should be posted.
- Storage temperature should not exceed 50 °C.

The classification requirements required for outdoor installation are given in the EN378 standard.

2. TECHNICAL INFORMATION

2.1. Technical Information

Table 4: Unit technical properties

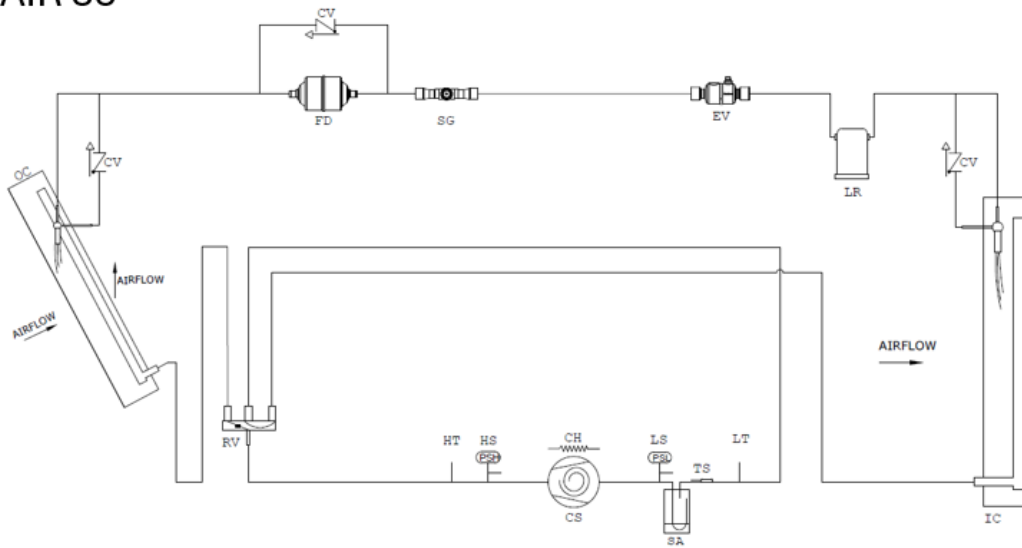
Models		AIR 35			AIR 45		
Performance Data	Unit	Summer	Summer (Max)	Winter	Summer	Summer (Max)	Winter
Outdoor Temperature (DB)	°C	35	40	5	35	40	5
Relative Humidity (RH)	%	40	30	83	40	30	83
Capacity	kW	38,39	36,36	34,06	46,21	45,74	42,71
Absorbed Power	kW	10,93	11,98	8,20	15,31	16,88	10,26
EER/COP	kW/kW	3,51 / -	3,03 / -	-/ 4,15	3,01 / -	2,70 / -	- / 4,16
Standard							
Refrigerant	-	R-454B					
Compressor Quantity	Qty.	1			2		
Compressor Type	-	Scroll					
Circuit Quantity	Qty.	1					
Capacity Control	-	1 step			2 steps		
Expansion Valve	-	Electronic					
Supply Fan							
Fan Type	-	EC Plug					
Airflow Rate	m³/h	6000			7250		
Fan Quantity	Qty.	1					
Static Pressure Drop	Pa	150					
Maximum Static Pressure Drop	Pa	500					
Filter		Supply					
Primary Filter	-	ISO Coarse 60% - (G4)					
Condenser Fan							
Fan Type	-	AC Axial			EC Axial		
Fan Quantity	Qty.	1					
Electrical Information							
Voltage	V	380~420 + N + PE					
Phase	ph	3					
Frequency	Hz	50					
Sound							
Sound Power Level	dBA	75			79		
Sound Level (10 meter)	dBA	47			51		
Dimensions							
Width	mm	1250			1050		
Length	mm	2270			2270		
Height	mm	2278			2245		
Weight	kg	735			750		
Structure							
Unit Casing	Oven Baked Powder Paint - 275 gr/m² Galvanized Steel						
Unit Insulation	25 mm thickness, 70 kg/m³ Rockwool						

*The values in the table are provided based on 100% fresh air supply.

2.2. Cooling Cycle

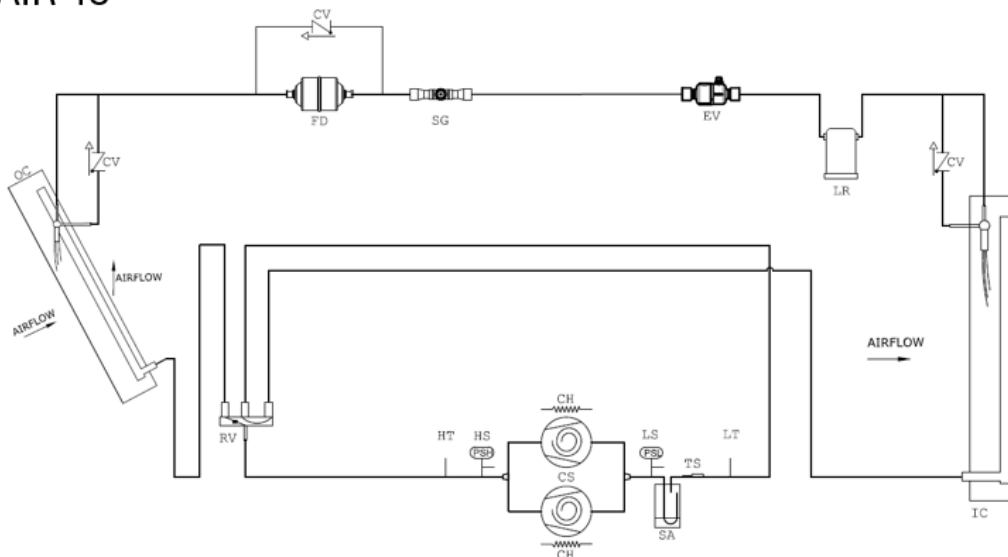
Figure 6: Unit Cooling Cycle

AIR 35



CH: CRANKCASE HEATER	OC: OUTDOOR COIL
CS: COMPRESSOR	FD: BI-FLOW FILTER DRIER
HS: HIGH PRESSURE SWITCH	SG: SIGHT GLASS
LS: LOW PRESSURE SWITCH	EV: ELECTRONIC EXPANSION VALVE
HT: HIGH PRESSURE TRANSMITTER	CV: CHECK VALVE
LT: LOW PRESSURE TRANSMITTER	LR: BI-FLOW LIQUID RECEIVER
TS: TEMPERATURE SENSOR	IC: INDOOR COIL
RV: 4-WAY REVERSING VALVE	SA: SUCTION ACCUMULATOR

AIR 45



CH: CRANKCASE HEATER	OC: OUTDOOR COIL
CS: COMPRESSOR	FD: BI-FLOW FILTER DRIER
HS: HIGH PRESSURE SWITCH	SG: SIGHT GLASS
LS: LOW PRESSURE SWITCH	EV: ELECTRONIC EXPANSION VALVE
HT: HIGH PRESSURE TRANSMITTER	CV: CHECK VALVE
LT: LOW PRESSURE TRANSMITTER	LR: BI-FLOW LIQUID RECEIVER
TS: TEMPERATURE SENSOR	IC: INDOOR COIL
RV: 4-WAY REVERSING VALVE	SA: SUCTION ACCUMULATOR

2.3. Equipment Information

Table 5: Equipment Information

Portable Air Conditioners		Standard	Optional
Filter	ISO Coarse 60% - (G4) Filter	*	
Heating	Heat Pump	*	
	Electric Heater		*
	Hot Water Coil		*
	Supply Temperature Sensor	*	
	Return Temperature Sensor	*	
	Outdoor Temperature Sensor	*	
	Supply Humidity Sensor		*
	Return Humidity Sensor		*
	Outdoor Humidity Sensor		*
	Pre Filter/Filter Clogging Alarm	*	
	Fan Differential Pressure Probe		*
	Fire Detector Input		*
	Remote Control Panel	*	
	Connectivity to Building Automation with ModBUS	*	
	System Integration with Lanwork or Bacnet Protocols		*
	Remote Access Capability (SNMP) with Modem over Ethernet or ModBUS		*
	Remote Connection Kit		*
	Data Log		*
	Daily - Weekly Schedule	*	
	Superheat Sensor	*	
	Low Pressure Switch	*	
	High Pressure Switch	*	
	Low Pressure Transmitter	*	
	High Pressure Transmitter	*	
	Condenser Fan Speed Control		*
Portable Air Conditioners		Standard	Optional
Supply – Return Fan	AC Plug Fan		*
	EC Plug Fan	*	
Cooling Equipment	Electronic Expansion	*	
	Inverter Compressor		*
	Hot Water Coil, Three Way Valve		*
	Stainless Inner Body		*
	Hydrophilic or Epoxy Coated Evaporator		*
	Hydrophilic or Epoxy Coated Condenser		*
Operation	Cooling	*	
	Heat Pump	*	
Main Equipment	Copper Tube Aluminum Finned Coil	*	
	Condenser Protection Grill	*	
	Axial Type Condenser Fan	*	
	Proportional Condenser Fan	*	
	Scroll Compressor	*	
	Asymmetrical Cooling Circuit		*
	Silent Models		*
	Compressor Crankcase Heater	*	

2.4. Dimensions and Weight

Figure 7: Unit dimension

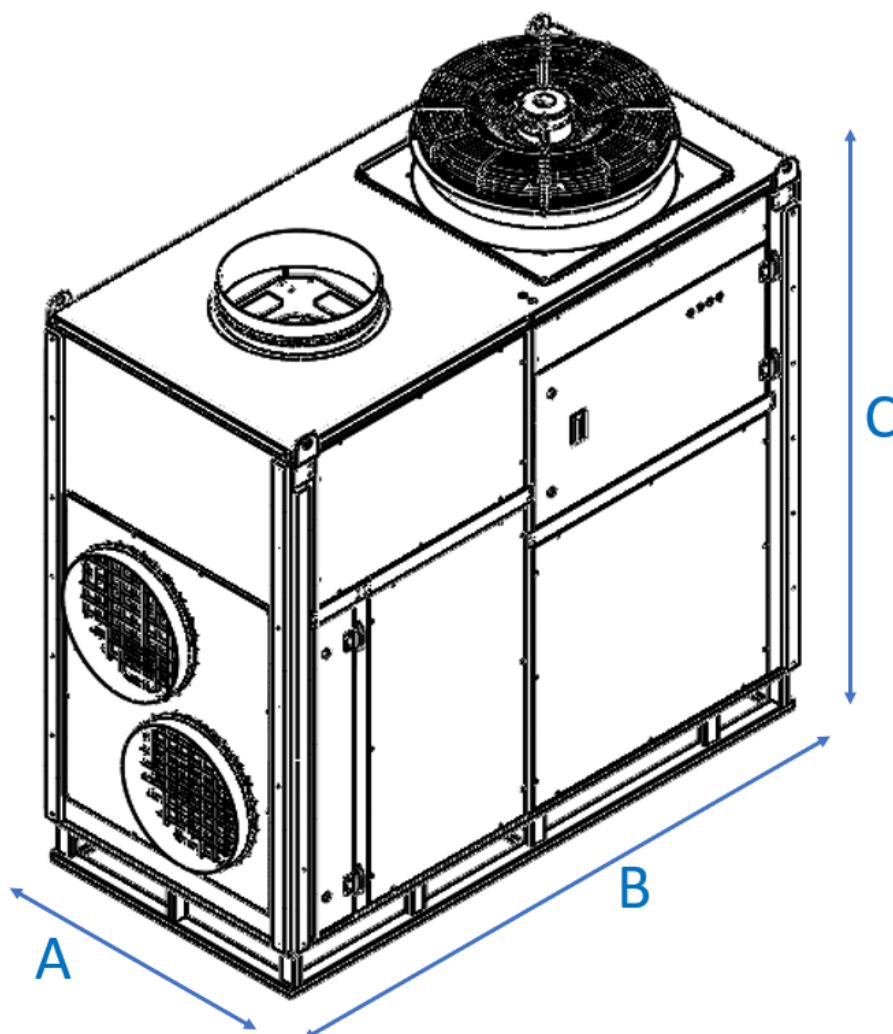


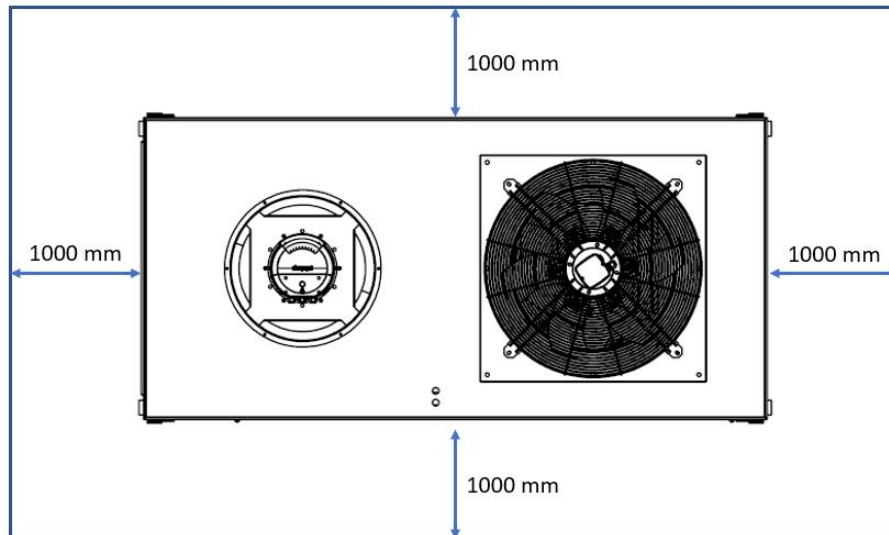
Table 6: Unit dimension and weight

Model	A (mm)	B (mm)	C (mm)	Weight (kg)
AIR 35	1250	2270	2278	735
AIR 45	1050	2270	2245	750

2.5. Minimum Maintenance Clearance

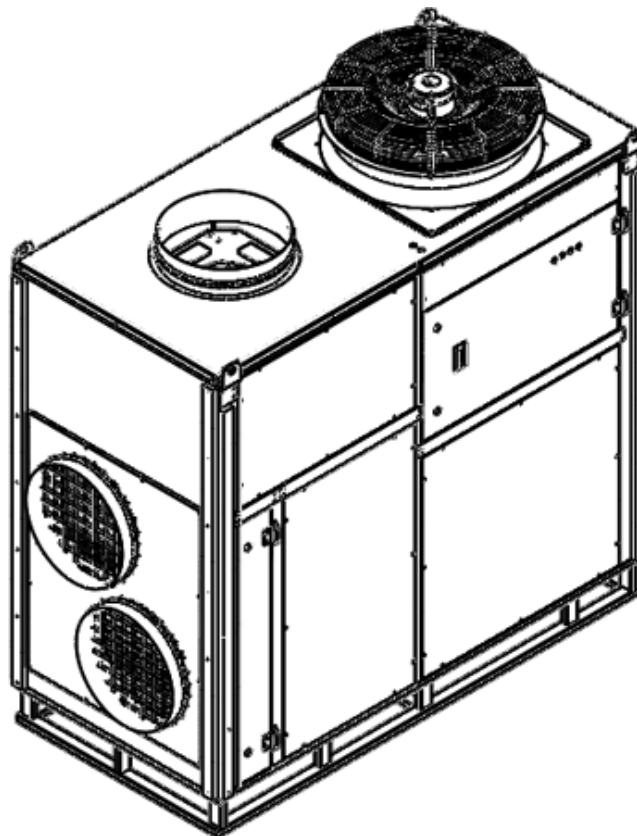
The minimum service clearance that should be left around the units are shown in Figure 4. Service and maintenance will be impossible if space is left below the minimum distance for each unit. Leave a minimum of service space around the unit.

Figure 8: Minimum service clearance



2.6. Unit Drawings

Figure 9: Unit isometric view



3. CHECK LIST

3.1. Delivery Options

- Following situations should be checked when a new unit is delivered. It is the buyer's responsibility to properly check the unit upon delivery to make sure it is firm and operational.
- Any Figure impairment such as scratches, dents, wrenches or bending should be checked
- Delivery and order details consistency should be checked
- Lifting and carrying process of the units should follow the instructions from that section of the manual.
- Ordered assembly accessories for installation should be checked to make sure they are in one piece and operational
- Helot and the transportation company should be notified in case of any error or damage to the unit from the transportation process in 48 hours with e-mail.
- The e-mail notifying the damage should include Figure documents such as pictures.
- Helot cannot be held responsible if the transportation damage is not sent in.

3.2. Before Installation

- There must be sufficient space in the place where the unit will be installed.
- The surface on which the unit will be installed must be strong and durable enough to support the weight of the unit.
- There should be no elements around the unit that may prevent it from working.
- The electrical power to be supplied to the unit must comply with the electrical characteristics of the unit.
- Drainage is required to drain the condensed water.
- For maintenance it must be located in accordance with the space and access drawings on page 26 in this manual.
- Installation of the unit may require different removal methods, which may vary depending on the installation. Pre-evaluation should be made in order to avoid a negative situation at the time of installation.
- The surface on which the unit will be mounted should be clean and there should be no factors that could obstruct the air flow.
- The unit must be kept away from sloping, bump, pit, etc. surfaces.
- Two units should not be installed close to each other. This may restrict the air flow to the condensers and the unit may not operate as desired.
- The unit should be arranged in such a way that it has the necessary space for full opening and closing of the unit doors, taking into account the installation and post-maintenance situations.

3.3. Connections

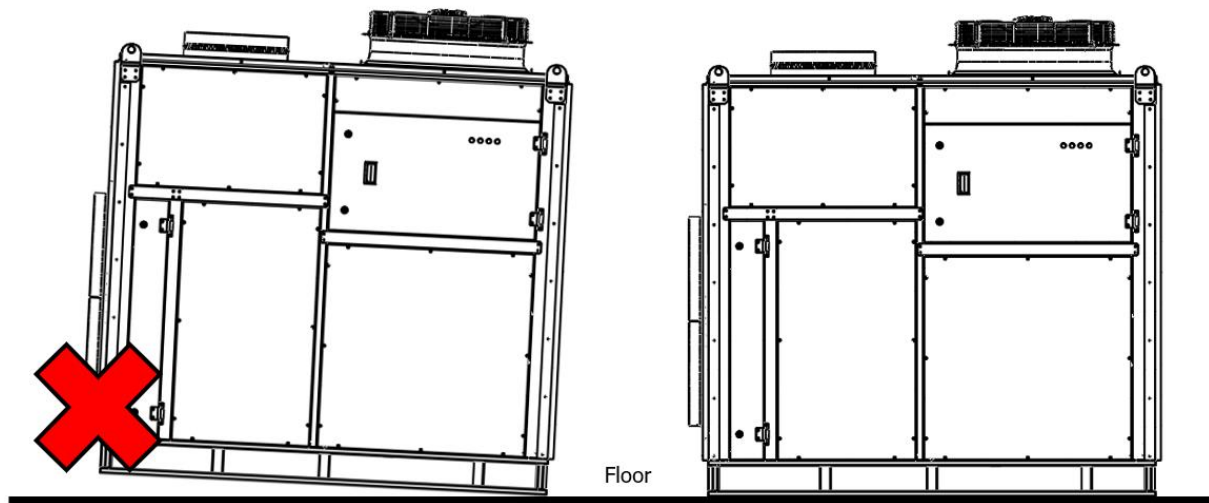
- It should be ensured that all blowing and suction channels are fixed to the unit and to each other.
- To avoid condensation problems, it should be ensured that all discharge ducts are insulated.
- Air duct, electrical connections, water connections etc. component weights must not be transferred to the unit. No extra load should be placed on the unit.

Note: The packaging protection must be removed before running the unit.

3.4. Before Running

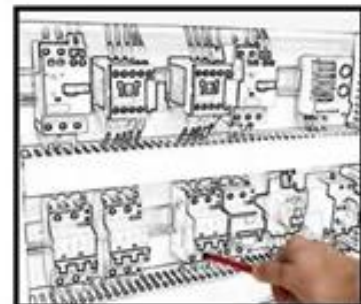
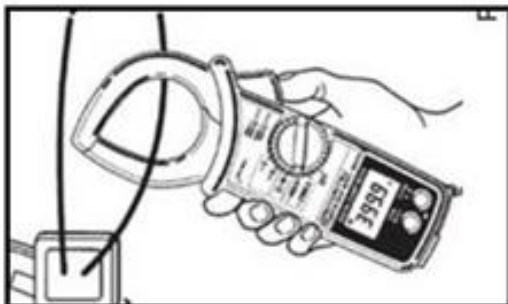
Check the unit is properly and levelly placed, and the cooling and electrical lines are correctly connected.

Figure 10: Placing the unit on the floor



Check that the electrical voltage and frequency are within the specified ranges. Make sure all electrical connections are properly tightened.

Figure 11: Electric connections check



4. CARRYING and STORAGE

4.1. Considerations During Transportation

- The unit should always be placed on even ground and anchored to a frame when transporting and the user should always act with caution.
- Damaging the unit by impacting it to any surface should be avoided at all costs. Any application that has a risk of damaging the product should not be carried out.
- The transportation should only be done by appropriate carrying equipment such as cranes and forklifts. Please see the correct and incorrect carrying methods below.
- Do not stand under or above the unit without any safety measures at any time.
- All applications should be conducted under the local law and regulations.
- Make sure the crane or the forklift has adequate capacity to carry the unit's weight.

4.2. Carrying with Crane or Crane-like Vehicles

- The carrying equipment should be placed in the holes located at the bottom of the unit.
- The top side should have a carrying console
- The unit should be hanged as shown in the Figure 8 after making sure it is safe to do so.
- The lifting belts should be tightened in steps and should be continuously checked if they are in the correct places.
- Highering the unit should be done slowly and the unit should be balanced at all times.
- Only authorized personnel should handle the crane and the forklift.
- The user is responsible for any damages that occur during the transportation unless there is Helot personnel present at the site.

The unit must be lifted using the spacer beams to prevent the straps from damaging the chassis. The length of the intermediate beams must be equal to the machine width.

4.3. Carrying with Forklift or Forklift-like Vehicle

- The forklift forks should be at least as long as the pallets themselves and they should be pushed all the way through
- The unit should be lifted slightly initially to make sure it is balanced.
- The unit should not be lifted on its short side. This may damage the unit.
- Lifting should be done with a forklift suitable for the size of the unit.
- The plastic cover of the unit should not be removed while the unit is being lifted.
- Care must be taken not to injure anyone while removing the forklift rails. When removing the forklift rails from the unit, it must be placed in a safe area.

4.4. Unpacking

- The carrying belts and ropes should not be cut.
- Cards, plastic covering and the wedges should not be removed
- The pallets or platforms below the unit should only be removed when the unit is lifted from the ground in a safe way.
- The unit should be placed to the area it will be used.
- User should check if there is any damage done to the unit.
- The removed packaging should only be discarded in accordance with the local law and regulation. It should be recycled if possible.

Figure 12: Carrying with Crane or Crane-like Vehicles

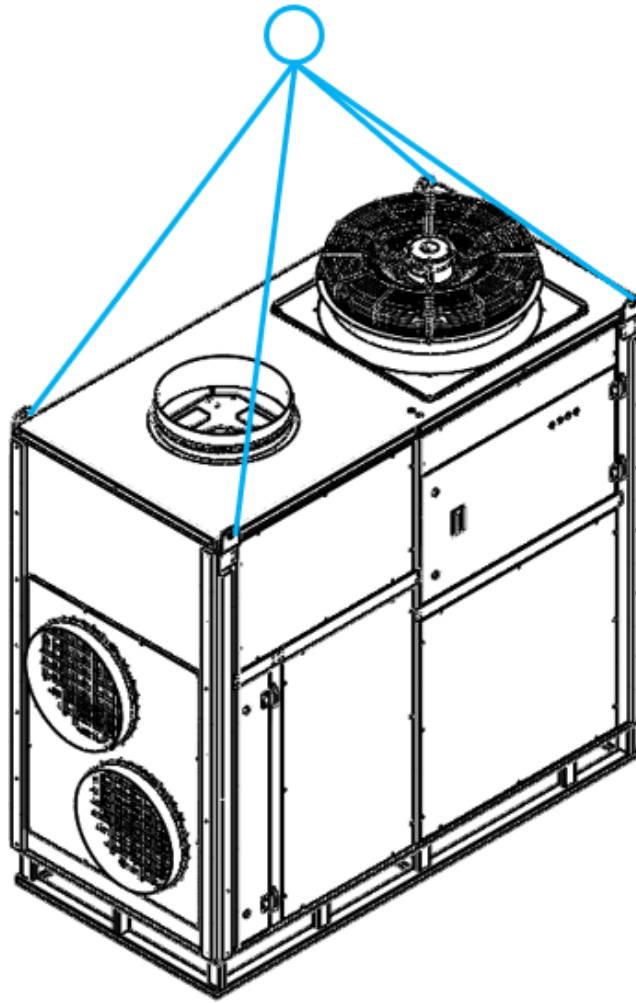
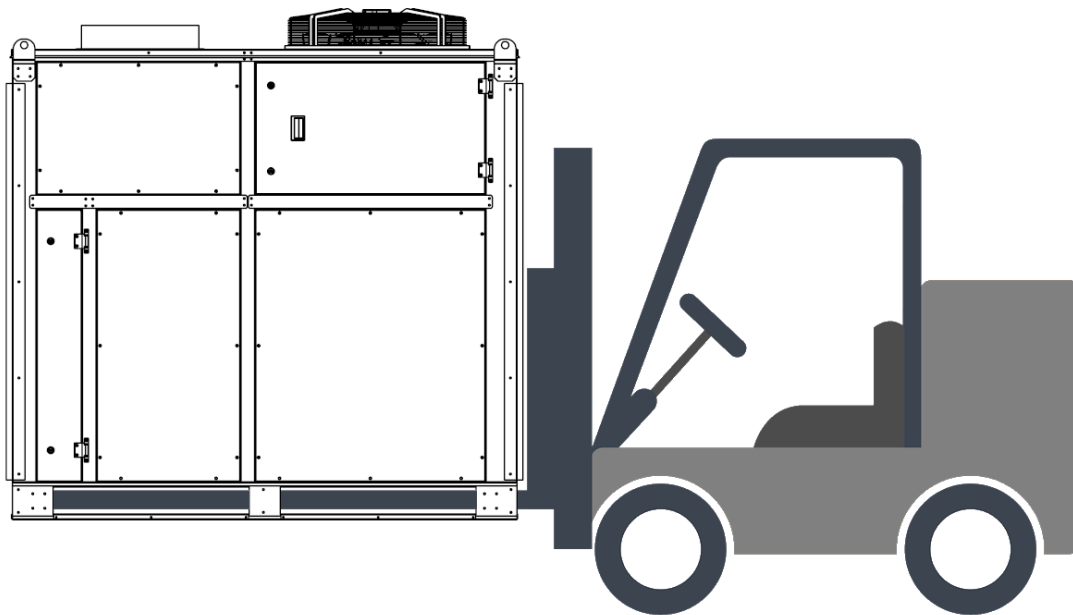


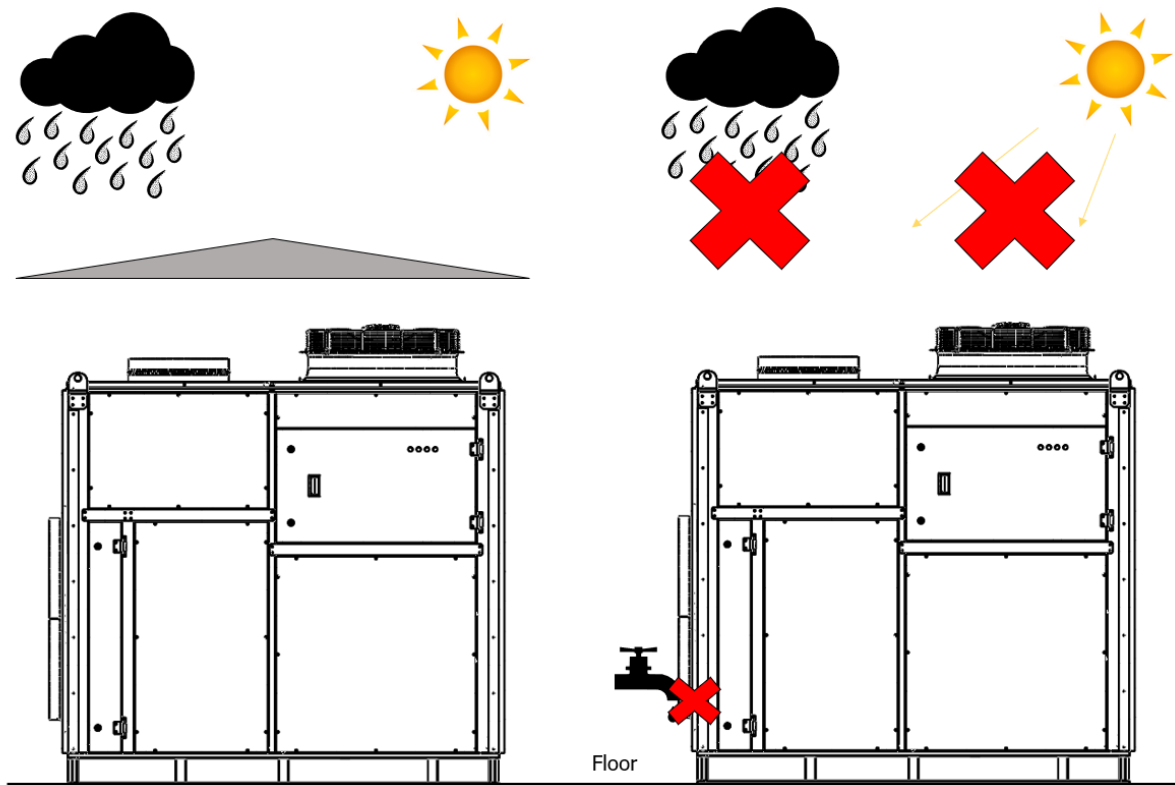
Figure 13: Carrying with Forklift or Forklift-like Vehicle



4.5. Considerations During Storage

- The unit is shipped with adequate packaging. When the unit received, any transport damage should be checked. If the unit is damaged or missing packaging, contact your transportation insurance company.
- The unit should be stored with its packaging until it is ready to be used.
- The unit should be protected against weather conditions such as rain, snow, sun, lightning etc and should be kept in a closed area while it is being stored for later use.
- The unit should be stored in a way that it will not be damaged and therefore should not be stored along with other equipment that can potentially damage it. The unit's storage space should be away from any industrial process.
- The unit should be stored on an even surface and should be kept away from dust, water and humidity.
- The unit is not stackable and no other equipment should be placed on top of it.
- Do not walk on the unit.

Figure 14: Correct and incorrect storage

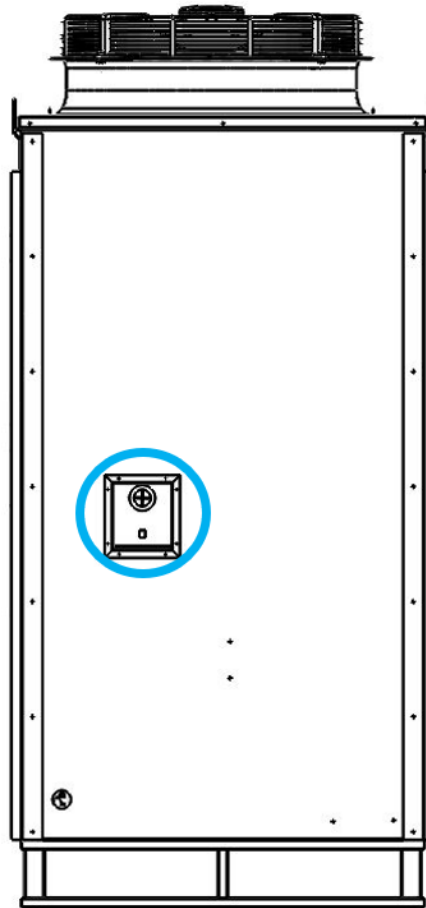


5. INSTALLATION

5.1. About Installation

Move the unit to the area to be installed. Make the electrical connection from the point shown in Image 17 below. Since the unit is plug-and-play, no external installation is required.

Figure 15: Electrical connection point



5.2. Electrical connection

- There should be no exposed wires.
- It should be checked that all connections are made properly.
- The grounding connection of the unit must be made.
- Cable cross section suitable for the electrical power consumption of the unit should be used.
- This connection must be made properly and completely.
- Main switch is turned on position and the unit will be energized

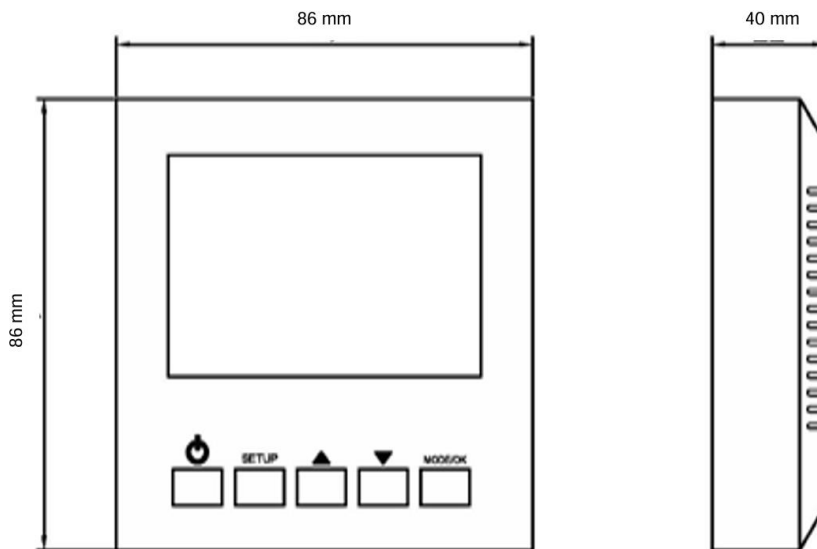
6. COMMISSIONING

6.1. Control Panel



Figure 16: Control panel

6.1.1. Control Panel Technical Dimensions



6.1.2. Control Panel Technical Specifications

Electrical Data	Supply	24 VAC/DC $\pm$ %10
	Power consumption	Max. 1VA
	Connection	1,5mm ² Terminal Connector
Function Data	Analog Output	0-10V Analog Output
Physical Data	Mounting Type	Wall Mount
	Dimensions	D86xG86xY40mm
	Weight	140 Gr
Environmental Data	Storage/Operating Temperature	0... +40°C/0... +70°C (Without Icing and Condensation)
	Protection Class	EN 60529 According to Standards IP20

Control Panel Functional Specifications:

- ❖ 3.2" Graphic Display
- ❖ User Key Lock
- ❖ Modbus Communication (Optional)
- ❖ Real Time Clock (Optional)
- ❖ Precise Control with PI
- ❖ Operating modes: Comfort, Economy and Protection, Minimum and maximum set point limitation
- ❖ Configurable User Parameters
- ❖ On/Off Heating/Cooling Valve Control
- ❖ DC 0...10 V output for actuator and modulating heating units
- ❖ Adjustable minimum and maximum air flow signal limit DC 0...10V
- ❖ Output Signal Conversion (Optional)
- ❖ Return Air Sensor Input (Ntc 10k)
- ❖ Window Contact (Universal Input that Can Be Used as External Sensor)
- ❖ Minimum And Maximum Set Temperature Limit

6.1.3. Panel Guide

(⏻)ON/OFF Button	It manages the ON/OFF functions of the unit.
(Mode/OK) Mode Button	Selects the operating mode of the device (Automatic Heating/Cooling and Fan, Manual Heating/Cooling and Fan mode). It is used as a confirmation button on device setting selection and mode screens.
(▲)Up Button	Increases Set Temperature in Auto mode.
(▼)Down Button	Decreases Set Temperature in Auto mode.
(SETUP) SETUP Button	It redirects to the User Settings Menu. Press and hold the SETUP button for 2 seconds to enter the User Setting Menu.
Key Lock Activation (SETUP+ ▼)	While the unit is on or off, press and hold the SETUP and DOWN buttons for 2 seconds at the same time. When you see "KEY LOCK ON" on the screen, the panel has switched to key lock. When the unit is on, there is a lock sign in the upper corner of the screen.
Key Lock Deactivation (SETUP+ ▼)	While the device is on or off, press and hold the SETUP and DOWN buttons for 2 seconds at the same time. When you see "KEY LOCK OFF" on the screen, the panel key lock is not active. Note: If the Controller is connected to the central system via Modbus Communication, the Key Lock can also be controlled from a remote point.

Table 7: Panel information

6.1.4. Weekly Schedule Configuration

- When the unit is on, press the **SETUP** button for 2 seconds on the panel.
- Select the Weekly Program option from the incoming settings menu and press the **MODE/OK** button.
- Select the day of the week from the Weekly Program screen with the **Up▲** or **Down▼** button and enter the start and end times of the device using the **MODE/OK** button. Proceed by pressing the **OK Button**.
- Note1: If the start time is after the end time, the device will remain off during the selected day.
- Note2: If the start time and end time are the same, the weekly program application will not work for the selected day.
- Note3: If the device is programmed weekly for the current day and is within the programmed working hours, the clock logo appears on the screen.
- Note4: If the device is programmed weekly for the current day and is within the programmed working hours, the clock logo stop logo appears on the screen.

6.1.5. Date and Hour Configuration

- While the device is on, press the **SETUP** button on the panel for 2 seconds.
- Select the Date Time Setting option from the incoming settings menu and press the **MODE/OK** button.
- Set the date and time using the **Up/Down** and **MODE/OK** buttons on the Date Time setting screen. Exit by pressing the **MODE/OK** button.

6.1.6. Language Configuration

- While the device is on, press the **SETUP** button on the panel for 2 seconds.
- Select the Language Selection option from the incoming settings menu and press the **MODE/OK** button.
- Set the language to Turkish or English from the Up or Down buttons on the Language Selection screen. Exit by pressing the **MODE/OK** button.

6.1.7. Display Brightness Configuration

- When the device is on, press the **SETUP** button for 2 seconds on the panel.
- Select the Light Intensity option from the incoming settings menu and press the **MODE/OK** button.
- Adjust the light intensity from the Up or Down buttons on the Light Intensity screen.
- Exit by pressing the **MODE/OK** button.

6.1.8. Screen Contrast Configuration

- When the unit is on, press the **SETUP** button for 2 seconds on the panel. From the incoming settings menu, highlight the Contrast Adjustment option and press the **MODE/OK** button. Adjust the light contrast value from the Up or Down buttons on the Contrast Adjustment screen.
- Exit by pressing the **MODE/OK** button.



ATTENTION: The technical personnel should do the changes in the Attention Service Setting Menu.

6.2. Service Configuration

6.2.1. Temperature Sensor Position Setting

- Press the **Up** and **Down** buttons at the same time for 3 seconds when the unit is in the off position on the panel. Select Control Temperature from the Service Menu. Select the position of the Sensor (Panel, Return Temperature) with the Up and Down buttons on the Control Temperature screen.
- Exit by pressing the **MODE/OK** button.

6.2.2. Defrost Service Setting

- When the unit is off, press the **Up ▲** and **Down ▼** buttons for 3 seconds on the panel. Select the Defrost Setting option from the Service Menu.
- Select the Defrost Minimum Temperature value from the Defrost Minimum Temperature screen with the **Up ▲** and **Down ▼** buttons. (If the temperature value measured from the evaporator temperature sensor remains below the Defrost Min. Temperature value, the device switches to defrost mode.) Press the **MODE/OK** button to proceed.
- Select the Defrost Maximum Temperature value from the Defrost Maximum Temperature screen with the **Up ▲** and **Down ▼** buttons. (When the device is in defrost mode, if the temperature value measured from the evaporator temperature sensor exceeds the Defrost Maximum Temperature value, the device will exit the defrost mode.) Press the **MODE/OK** button to proceed.

6.2.3. Outdoor Temperature Display

- When the unit is off, press the **Up ▲** and **Down ▼** buttons on the panel for 3 seconds.
- Select Outside Temperature from the Service Menu. Select On or Off from the Outdoor Temperature screen with the Up and Down buttons. Exit by pressing the **MODE/OK** button.
- Note: If the Outdoor Temperature Indicator is On, NTC 10k sensor should be installed at the outside temperature sensor input of the control card.

6.2.4. Minimum Set Temperature Setting

- When the device is off, press the **Up ▲** and **Down ▼** buttons on the panel for 3 seconds.
- Select the Set Minimum option from the Service Menu. Set the minimum set temperature limit with the Up and Down buttons on the Set Minimum screen. Exit by pressing the **MODE/OK** button.

6.2.5. Maximum Set Temperature Setting

- When the unit is off, press the Up and Down buttons on the Panel for 3 seconds.
- Select the Set Maximum option from the Service Menu. Set the set temperature maximum limit with the Up and Down buttons on the Set Maximum screen. Exit by pressing the **MODE/OK** button.

6.2.6. Automatic Mode Minimum Fan Speed Temperature

- When the unit is off, press the Up and Down buttons on the Panel for 3 seconds.
- Select FanAutoMinLimit from the Service Menu. Select the minimum fan speed value from the FanAutoMinLimit screen with the Up and Down buttons. Exit by pressing MODE/OK button.

6.2.7. Condenser Set Temperature Value Setting

- When the unit is off, press the Up and Down buttons on the Panel for 3 seconds.
- Select the Compressor Temperature Set option from the Service Menu. While the unit is in cooling mode, select the operating temperature set value of the condenser fan from the Compressor Temperature Set screen with the Up and Down buttons. Exit by pressing the MODE/OK button.
- While the unit is in cooling mode, select the operating temperature set value of the condenser fan from the Compressor Temperature Set screen with the Up and Down buttons. Exit by pressing the MODE/OK button.

6.2.8. Compressor Start Time Setting

- When the unit is off, press the Up and Down buttons on the Panel for 3 seconds.
- Select Compressor Start Time from the Service Menu. Select the compressor start time from the Compressor Start Time screen with the Up and Down buttons. Exit by pressing the MODE/OK button.

6.2.9. Compressor Stop Time Setting

- When the unit is off, press the Up and Down buttons on the Panel for 3 seconds.
- Select Compressor Stop Time from the Service Menu. Select the stop time of the compressor from the Compressor Stop Time screen with the Up and Down buttons. Exit by pressing the MODE/OK button.

6.2.10. Panel Communication Setting

- When the unit is off, press the **Up ▲** and **Down ▼** buttons on the Panel for 3 seconds. Select the Panel Setting option from the Service Menu. Select the first Modbus ID from the Panel Setup screen with the **Up ▲** and **Down ▼** buttons.
- Press the **MODE/OK** button to switch to the Modbus Baudrate Value Screen. On the Modbus Baudrate screen, select the Baudrate (1200Bps, 2400Bps, 4800Bps, 9600Bps, 19200Bps, 38400Bps). Exit by pressing the **MODE/OK** button.
- Note1: Panel communication works with Modbus RTU protocol over RS485 Line. In desired applications, it can be operated with devices such as PC, Scada, PLC, HMI on the Panel communication port.
- Note2: Communication Settings: Data Bits:8, None Parity Stop Bit 1.

6.2.11. BMS Modbus Communication Setting

- When the unit is off, press the **Up ▲** and **Down ▼** buttons on the Panel for 3 seconds.
- Select Modbus Setting option from Service Menu. Select the first Modbus ID from the COM2 Setup screen with the **Up ▲** and **Down ▼** buttons. Press the **MODE/OK** button to switch to

the Modbus Baudrate Value Screen. On the Modbus Baudrate screen, select the Baudrate (1200Bps, 2400Bps, 4800Bps, 9600Bps, 19200Bps, 38400Bps). Exit by pressing the MODE/OK button.

- Note1: Building automation communication works with Modbus RTU protocol over RS485 Line. Panel communication port in desired applications (It can be operated with devices such as PC, Scada, PLC, HMI.)
- Note2: Communication Settings: Data Bits:8, None Parity Stop Bit 1.

6.2.12. Panel Factory Reset

- When the unit is off, press the **Up ▲** and **Down ▼** buttons on the Panel for 3 seconds. Select Factory Reset from the Service Menu.
- From the Factory Reset screen, select the "ON" option with the Up and Down buttons. Exit by pressing the **MODE/OK** button.
- The Room Panel will return to the Factory Settings.
- Panel Factory Settings:
 - Light Intensity: 50%
 - Contrast Adjustment: 50%
 - Panel Communication: Modbus ID: 1
 - Modbus Baudrate: 9600 Bps.

6.3. Modbus List

This Modbus Register List is for the CPX:139 Rooftop control system. The controller is suitable for devices up to version v1.1. If the connected device is above version v1.1, please contact Imbat. Other Modbus parameters Data bit 8, Data parity bit none (N), Stop bit 1 cannot be changed. All Modbus parameters are holding register. All Modbus parameters are Signed Integer16.

Parameter Set: PartNo: 111; 1

Password: 58

Table 8: Modbus list

PLC Register				
Device General Control Values				
40001	0	Unit On/Off Variable	0...1	0: OFF 1: ON
40002	1	Unit Set Temperature	0...999	280 means 28 °C.
40003	2	Unit Condition Mode	0...5	0: Fan Mod 1: Manual Heating 2: Manual Cooling 3: Automatic Heating 4: Automatic Cooling 5: Fully Automatic Mode
Proportional Controlled Systems				
40004	3	Aspirator Fan Manual Speed Value	1...3	
40005	4	Ventilator Fan Manual Speed Value	1...3	
Device Temperature Registers				
40006	5	Application Temperature	-400...999	280 means 28 °C.
40007	6	Panel Ambient Temperature Value	-400...999	Temperature Value on the Panel.
40008	7	Unit Return Air Temperature Value	-400...999	It shows the temperature of return air in the place.
40009	8	Unit Supply Air Temperature Value	-400...999	If the unit does not have a current temperature value, the temperature value is displayed as "32767".
40013	12	Kondenser Temperature Value	-400...999	
Instant Device Input Status Display Registers				
40015	14	Digital Input Values 1	0...1	0. Bit ==> BMS Input 1. Bit ==> Boost Unput 2. Bit ==> Aspirator Fan Air Flow Info 3. Bit ==> Ventilator Fan Air Flow Info 4. Bit ==> Filter 1 Pollution Info 5. Bit ==> Filter 2 Pollution Info 6. Bit ==> Electric Heater Safety Thermostat Info 7. Bit ==> Aspirator Fan Thermal Failure Information 8. Bit ==> Ventilator Fan Thermal Failure Information 9. Bit ==> Emergency Stop Button Info 10. Bit ==> Low Pressure Info 11. Bit ==> High Pressure Info

				12. Bit ==> Fire Info 13. Bit ==> Frozen Thermostat Info 14. Bit ==> Compressor Thermal Failure Information 15. Bit ==> Phase Failure Info
40016	15	Digital Input Values 2	0...1	0.Bit ==> Heat Exchanger Frozen Thermostat 1.Bit ==> VRF Failure 2.Bit ==> VRF Defrost 3.Bit ==> Filter 3 Pollution Info 4.Bit ==> Unit Door Open Info 5.Bit ==> Damper Does Not Work Properly Info 6. Bit==> Humidifier Failure Info
Digital Output Instant Status Registers				
40044	43	Digital Output 1	0...1	0==> Fresh Air/Return Air Damper Motor 1==> Bypass Damper Motor 2==> Heating/Cooling Coil Valve Output 3==> Heating Coil On/Off Output 4==> Cooling On/Off Coil Output 5==> Compressor Out 6==> Compressor 4-way Valve Output 7==> Unit Is Running Info Output 8==> Device Fault Information Output 9==> Unit Warning Info Output 10==> Aspirator Fan Run Output 11==> Ventilator Fan Run Output 12==> VRF On/Off Output 13==> VRF Heating Output 14==> VRF Cooling Output 15==> Rotary Output
40045	44	Digital Output 2	0...1	0==> Humidifier Output
Fan Heater Status Indicator Register				
40062	61	Ventilator Fan Instant Value	0...3	
40620	619	Kondenser Fan Instant Control Value	0...1000	
Instant Unit Equipment Status Indicator Registers				
40066	65	Ventilator Fan Analog Output Value	0...1000	
40067	66	Proportional Heating Valve Output Value	0...1000	
40068	67	Proportional Cooling Valve Output Value	0...1000	
40069	68	Proportional Heating/Cooling Valve Output Value	0...1000	
40070	69	Proportional Bypass Damper Output Value	0...1000	
40071	70	Mixture Damper Output Value	0...1000	
Weekly Schedule Time Adjustment Parameter Registers				
40073	72	Sunday Start; Hour:Minute	0...2359	The unit operates between the start and end times in a day and
40074	73	Sunday Stop; Hour:Minute	0...2359	
40075	74	Monday Start; Hour:Minute	0...2359	

40076	75	Monday Stop Hour:Minute	0...2359	the device switches off outside these hours. If the start and end times are equal, the device works continuously that day. If the start time is more than the end time, the device is completely turned off on the set day. If the adjusted parameter is 1210, Hour:Minute is set as 12:10.
40077	76	Tuesday Start; Hour:Minute	0...2359	
40078	77	Tuesday Stop; Hour:Minute	0...2359	
40079	78	Wednesday Start; Hour:Minute	0...2359	
40080	79	Wednesday Stop; Hour:Minute	0...2359	
40081	80	Thursday Start; Hour:Minute	0...2359	
40082	81	Thursday Stop; Hour:Minute	0...2359	
40083	82	Friday Start; Hour:Minute	0...2359	
40084	83	Friday Stop; Hour:Minute	0...2359	
40085	84	Saturday Start; Hour:Minute	0...2359	
40086	85	Saturday Start; Hour:Minute	0...2359	
40087	86	Current Day	1...31	
40088	87	Current Month	1...12	
40089	88	Current Year	0...99	
40090	89	Current Hour	0...23	
40091	90	Current Minute	0...59	
40092	91	Current Seconds	0...59	
40093	92	Current Day of the Week	1...7	1: Sunday 2: Monday 3: Tuesday 4: Wednesday 5: Thursday 6: Friday 7: Saturday
40094	93	Current Weekly Schedule Status	0...2	0: No Weekly Schedule 1: Weekly schedule set and unit is running. 2: Weekly schedule set and unit is not running.
Menu Setting Parameters				
40096	95	Alarm Value	0...65.535	
40097	96	Warning Value	0...65.535	
Fault Group Information Parameters				
40052	51	Defective Compressor No	0...4	It is the place where the information of the faulty compressor group and sensor group is given. For example, if compressor 2 is faulty, parameter 51 will be 2.
40053	52	Defective Sensor No	0...4	
40054	53	Low Pressure Defective Group No	0...4	
40055	54	Low Pressure Defective Group No	0...4	
Key Lock				
40104	103	Key Lock	0...1	0: Key Lock OFF 1: Key Lock ON
Failure Parameters				
40105	104	Very Important Failure	0...1	0: No Failure 1: The unit has completely stopped due to an error.
40106	105	Compressor Failure	0...1	0: No Compressor Failure 1: Compressor has stopped due to error.
Password Parameters				
40111	110	Service Password	0...9999	This is the password that must be entered in order to set the parameters with Parameter Number After 120. The password for V1.0 Motherboard is "58".
Building Automation Modbus-RTU Parameters				
40112	111	Modbus ID	1...254	
40113	112	Modbus Baud rate	0...6	0:1200Bps

				1:2400Bps 2:4800Bps 3:9600Bps 4:19200Bps 5:38400Bps 6:57600Bps
Universal Input Display Parameters				
40182	181	No. 1 Universal Input Status	-400...999	If the device's universal input is selected one of the device's digital inputs, the Universal input value appears as 0 or 1.
40183	182	No. 2 Universal Input Status	-400...999	
40184	183	No. 3 Universal Input Status	-400...999	
40185	184	No. 4 Universal Input Status	-400...999	
40186	185	No. 5 Universal Input Status	-400...999	
40187	186	No. 6 Universal Input Status	-400...999	
40188	187	No. 7 Universal Input Status	-400...999	If the device's universal input is one of the device's temperature inputs, the Universal input value appears as a temperature value between -400 and 999.
40189	188	No. 8 Universal Input Status	-400...999	
40190	189	No. 9 Universal Input Status	-400...999	
40191	190	No. 10 Universal Input Status	-400...999	
40192	191	No. 11 Universal Input Status	-400...999	
40193	192	No. 12 Universal Input Status	-400...999	
40194	193	No. 13 Universal Input Status	-400...999	
40195	194	No. 14 Universal Input Status	-400...999	
40196	195	No. 15 Universal Input Status	-400...999	
40197	196	No. 16 Universal Input Status	-400...999	
40198	197	No. 17 Universal Input Status	-400...999	
40199	198	No. 18 Universal Input Status	-400...999	
40200	199	No. 19 Universal Input Status	-400...999	
Digital Output Display Parameters				
40202	201	Digital Output 1 Status	0...1	Relay Output Open ==> Output:0
40203	202	Digital Output 2 Status		
40204	203	Digital Output 3 Status		
40205	204	Digital Output 4 Status		
40206	205	Digital Output 5 Status		
40207	206	Digital Output 6 Status		
40208	207	Digital Output 7 Status		Relay Output Close ==> Output:1
40209	208	Digital Output 8 Status		
40210	209	Digital Output 9 Status		
40211	210	Digital Output 10 Status		
40212	211	Digital Output 11 Status		
40213	212	Digital Output 12 Status		
40214	213	Digital Output 13 Status		
40215	214	Digital Output 14 Status		
40216	215	Digital Output 15 Status		
40217	216	Digital Output 16 Status		
40218	217	Digital Output 17 Status		
40219	218	Digital Output 18 Status		
40220	219	Digital Output 19 Status		
Analog Input Display Parameters				
40222	221	No. 1 Analog Input Status	-9999...9999	
40223	222	No. 2 Analog Input Status		
40224	223	No. 3 Analog Input Status		
40225	224	No. 4 Analog Input Status		
40226	225	No. 5 Analog Input Status		
40227	226	No. 6 Analog Input Status		
40228	227	No. 7 Analog Input Status		
40229	228	No. 8 Analog Input Status		

Analog Output Display Parameters				
40232	231	No. 1 Analog Output Status	0...1000	
40233	232	No. 2 Analog Output Status		
40234	233	No. 3 Analog Output Status		
40235	234	No. 4 Analog Output Status		
40236	235	No. 5 Analog Output Status		
40237	236	No. 6 Analog Output Status		
40238	237	No. 7 Analog Output Status		
40239	238	No. 8 Analog Output Status		

7. MAINTENANCE

7.1. Information for Using the Unit

Helot air conditioners are delivered as tested. The settings of your units should not be interfered with other than authorized service. Otherwise, it voids the warranty. Cooling circuit elements and automatic control elements in the units should not be disabled in any way. Electrical connections and electrical supply voltage should be checked frequently. Other loads should not be added to the same installation.

The electrical control panel of the unit must be dust-free. The contact points on the panel should be cleaned against oxidation, the current settings of the thermal relay and the current values drawn by the compressors should be checked. Changes in current may be a precursor to many faults. If the thermal and fuses of the unit blow frequently, the authorized technical service should be called immediately.

If the condenser of your unit is air cooled, it should be cleaned with compressed air or water at regular intervals according to the dustiness rate. If it is a water-cooled condenser, the condenser pressure loss should be checked and sufficient water flow should be provided.

We recommend that the maintenance of the unit is carried out periodically by HELOT authorized services.

7.2. Condenser Cleaning

Condensers are checked for pollution once a week. If there is dust and pollution, the condensers should be cleaned with the help of compressed air.

7.3. Compressor Oil Check

Compressor oil level is checked through the sight glass on the compressor crankcase once a week. If the oil level is below the minimum, the unit is turned off by pressing the on/off button of the digital display. If there is a problem, contact the authorized service.

For oil reinforcement, compressor and discharge valves are closed completely. The gas in the compressor is discharged by opening the service valves on the suction and discharge valves. The vacuum pump is connected to the service valve on the discharge valve. The service valve on the suction valve, on the other hand, is dipped into a pre-filled compressor oil (polyester) in a separate container by connecting a separate hose. By operating the vacuum pump, the oil is filled into the compressor crankcase. Sight glass is observed during filling. When the level approaches the maximum, the oil filling process is terminated. The inside of the compressor is vacuumed. After the service valves are closed, the suction and discharge valves are opened fully. After the process is completed, the operating instruction is executed.

7.4. Leakage Check

Welding points in the cooling installation are connected to each other with bolts or unions and plastered one by one with soap or detergent foam. If moving air bubbles form on the foam, there is a leak at that point. If the leak is in the places that are connected to each other with a union or bolt, the leak is removed by tightening the bolts and unions, and the test is repeated. If the leakage is at the source points, the gas of the unit is discharged by opening the service valves on the suction and discharge valves. Leaky areas are welded with oxyacetylene welding by using suitable welding wire. The leak test is repeated by pressing nitrogen gas at the service valves on the suction and discharge valves to the

system. If there are no other leaks, the system is vacuumed. The refrigerant gas is charged to the unit with the help of the same valves. After the process is completed, the operating instruction is executed.

7.5. Fan Check

If the fans are not working even though there is power, the unit is turned off. Check if there is anything attached to the fan blades. If nothing is blocking it, the fan motor junction box opens. The electrical connections are removed and the resistance at the winding ends is measured with the help of an ohmmeter. If there is no resistance, the fan motor has burned out and the fan motor winding must be rewound. The fan can be replaced with a new one. After the process is completed and the problem is fixed, the operating instruction is executed.

7.6. Prob Check

Shut down the unit power. Defective probe is removed. It is checked whether it shows resistance by connecting an ohmmeter to the probe cable ends. If it does not show resistance, the probe is replaced with a new one. After the process is completed and the problem is fixed, the operating instruction is executed.

7.7. Refrigerant Check

Suction and discharge pressures should be checked while the unit is in operation. If the pressure drops below 5 bar in the suction line and the pressure in the discharge line exceeds 35 bar, the Helot service team should be informed. The condition of the fluid should be observed by looking through the sight glass. If there is foaming, there is a lack of refrigerant. Refrigerant must be replenished to the unit. Contact the authorized service.

7.8. Filter Cleaning

During regular maintenance, the filter should be removed, cleaned and reinserted. Filter cleaning should be done with compressed air.

8. ERRORS and WARNINGS

Table 9: Error list

Value	Binary	Description
E1	0000.0000.0000.0001	Aspirator Fan Error
E2	0000.0000.0000.0010	Ventilator Fan Error
E4	0000.0000.0000.0100	Electric Heater Error
E8	0000.0000.0000.1000	Aspirator Air Flow Error
E16	0000.0000.0001.0000	Ventilator Air Flow Error
E32	0000.0000.0010.0000	Compressor Error
E64	0000.0000.0100.0000	Low Pressure Error
E128	0000.0000.1000.0000	High Pressure Error
E256	0000.0001.0000.0000	Fire Alarm
E512	0000.0010.0000.0000	Phase Error
E1024	0000.0100.0000.0000	VRF Error
E2048	0000.1000.0000.0000	Frozen Coil Error
E4096	0001.0000.0000.0000	Damper Error
E8192	0010.0000.0000.0000	Sensor Error
E16384	0100.0000.0000.0000	Condenser Fan Error

Table 10: Warning List

Value	Binary	Description
W1	0000.0000.0000.0001	Closed by Building Automation System
W2	0000.0000.0000.0010	Boost Mode
W4	0000.0000.0000.0100	Filter 1 Pollution
W8	0000.0000.0000.1000	Filter 2 Pollution
W16	0000.0000.0001.0000	Defrost Modu
W32	0000.0000.0010.0000	Frozen Thermostat
W64	0000.0000.0100.0000	Emergency
W128	0000.0000.1000.0000	VRF Defrost Warning
W256	0000.0001.0000.0000	Filter 3 Pollution
W512	0000.0010.0000.0001	Panel Door Open
W1024	0000.0100.0000.0000	Humidifier Error

9. SPARE PART LIST

Spare parts list of cooling components is in the annex of the user manual.

The electrical diagram and the electrical component list are in the annex of the user manual.